



GE Medical Systems

Technical Publication

Direction 2300338-100

Revision 0

**GE Medical Systems
SGI Octane 2 FRU, Replacement
Instructions**

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- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
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- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
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WARNUNG

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- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A' CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	06/12/01	Initial Release

List of Effected Pages

PAGES	REVISION	PAGES	REVISION
1 through 24	0		

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Octane 2 - FRU Replacement Procedure

Section 1.0 Before you Begin

1.1 “About” The Octane Computer FRU

The Octane computer field replaceable unit (FRU) is shipped configured to re-use components from the computer it replaces. The computer is shipped minus the following components:

- Disk Drives
- PCI Module and PCI boards
- Memory Modules
- System ID Module

You will remove these components from the computer being replaced and re-install them in your replacement computer. All of the above components are also available as individual FRUs.

1.2 Avoiding Equipment Damage and Personal Injury

Avoid damage to components and personal injury by taking the following precautions:

- Do not touch or clean the compression connector.
- Place the protective cap over the compression connector, if available.
- Lay all modules and board on antistatic surfaces only.
- Place all modules and board in antistatic bags.
- Wear a grounded ESD wrist strap in good working order.
- Maintain a clean work environment.
- Tag and lockout power.
- Wait five minutes after power is off before you continue. Let the system module cool.

1.3 Disk Drives

Disk drives are packaged in special carriers and require no assembly or disassembly. The primary system drive is placed in the bottom bay and the (image) disk in the middle drive bay. No jumpers are required because the computer's hardware automatically assigns SCSI IDs.

1.4 The ID Module

The system ID module is unique to your CT system. The System ID Module contains the computer's unique “Ethernet Address” number. If you loose the ID module, all of the Software Options you have installed will not be available for use. System software is locked to this unique ID during installation.

1.5 Replacement Process Overview

The replacement process is quite simple. In order to install the replacement computer, you must re-use components from the computer being replaced. First, remove the components specified in Section 1.1. from the computer in the console. Next, re-install them in the replacement computer that you have received. The flow chart in Figure 1 illustrates this process.

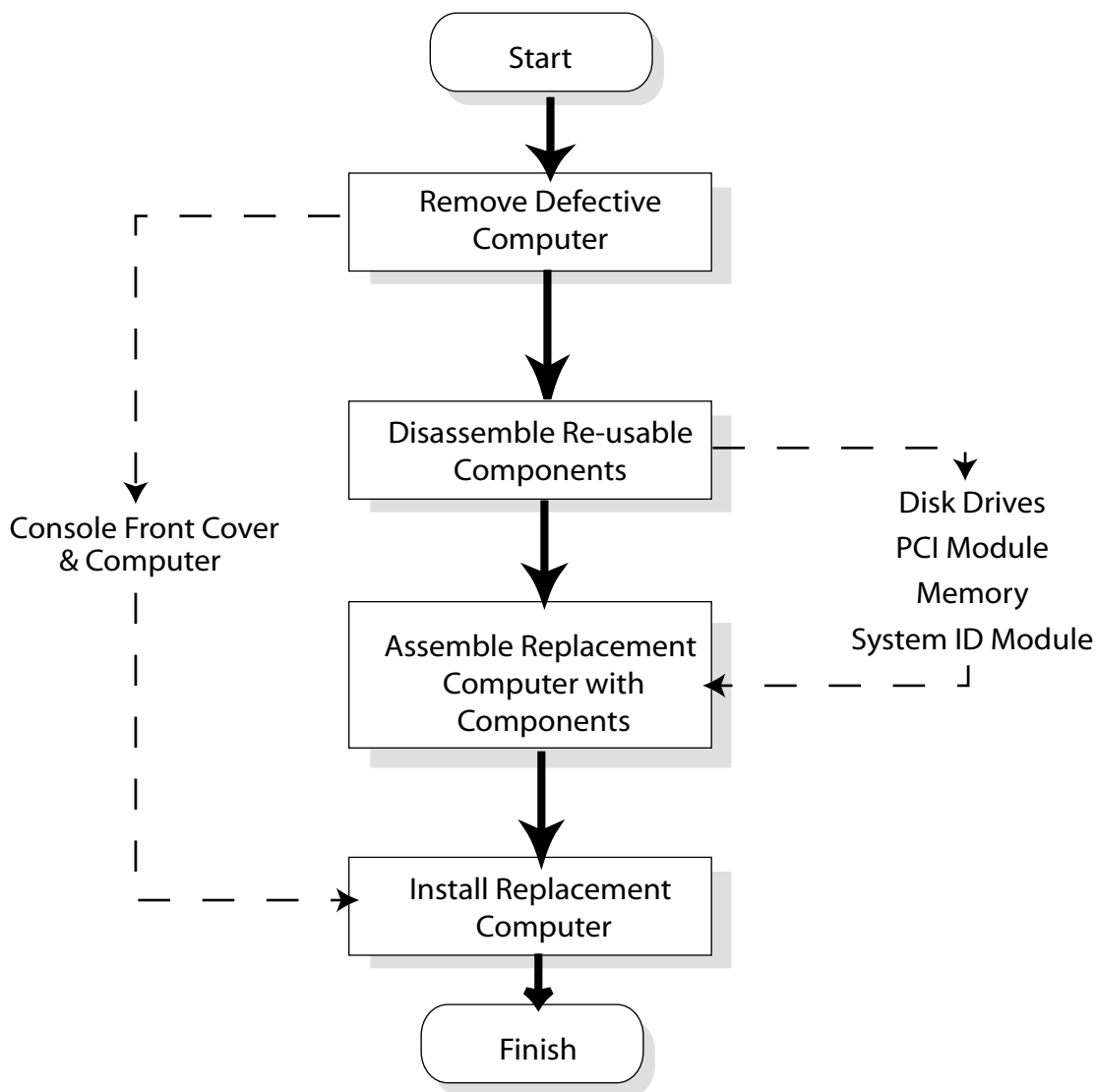


Figure 1 - Replacement Process

1.6 Supporting Documentation

For additional replacement details or help on troubleshooting problems that might occur, consult the console chapter in your CT System Service Manual that came with your product.

Section 2.0

The Replacement Process



NOTICE
Potential for
Equipment
Damage

Electronic devices are extremely sensitive to ESD damage. Always do the following:

- Wear a grounded ESD wrist strap.
- Place parts on antistatic surfaces.
- Transport parts in antistatic bags.

2.1 Remove Defective Computer

2.1.1 Console Front Cover

- 1.) From the service desktop, shut down the system.
- 2.) Tag and lockout system power.
- 3.) Turn off console power.
- 4.) Remove the console's front cover. Use a 4mm "allen" wrench to rotate each pull-ring counterclockwise.

2.1.2 Computer

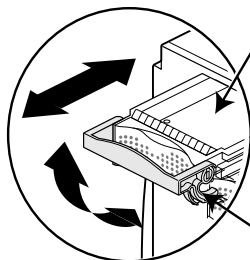
- 1.) Pull computer tray out.
- 2.) Disconnect the cables attached to the back of the computer.
- 3.) Lift the computer out and place on a stable and clean work surface.

2.2 Disassemble Re-usable Components

2.2.1 Disk Drives

- 1.) Remove the front bezel (cover) from the computer. Press both bezel release buttons simultaneously to release the cover.
- 2.) Tilt the bezel toward you and lift up to remove.
- 3.) Grasp a drive handle and lift it to a horizontal position.

Slide Drive "In" to install
Slide Drive "Out" to remove



Rotate handle Down to lock Disk
Rotate handle UP to release Disk

Figure 2 - Removing & installing Disk Drives

- 4.) Applying gentle and steady pressure, pull the drives out.
- 5.) Place each drive on an antistatic surface or in an approved antistatic bag.
- 6.) Re-install the front cover.

2.2.2 PCI Module

- 1.) Loosen the two captive screws (see [Figure 3](#)).
- 2.) Pull out the release lever along the bottom of the module (see [Figure 3](#)).

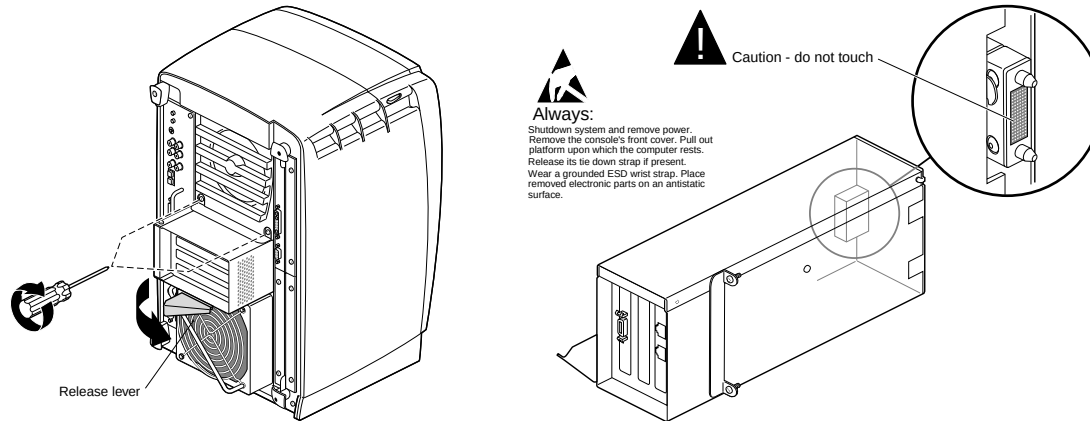


Figure 3 - Removing the PCI Module

- 3.) Slide out the module, taking care not to allow the compression connector to touch anything. Cap the compression connector, once the module is resting on an antistatic surface.

2.2.3 Memory



CAUTION
Potential for
Personal Injury

Allow computer components to cool. Wait 5 minutes after shutting off power before handling computer components

- 1.) Remove the system module.
 - a.) Loosen the captive screw in each sliding handles.
 - b.) Pull both handles at the same time until they are fully extended ([Figure 4](#))

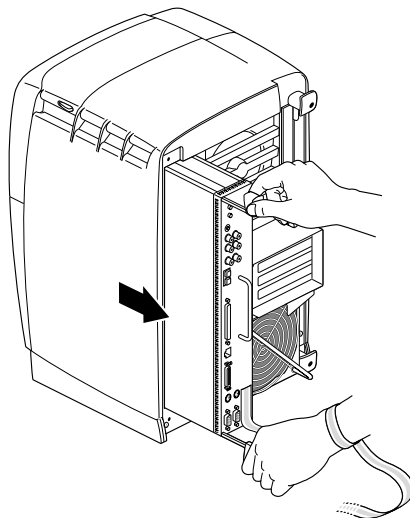


Figure 4 - System Module

- c.) Grasp the module handle with your left hand and place your right hand against the top of the computer's back. Pull the module out without allowing the delicate connectors on its back edge to touch anything.
 - d.) Place the system module on an antistatic surface component side up.
- 2.) Remove the Memory Modules
- a.) Locate the memory on the System Module and unlatch them. Press down on the latch, as shown in [Figure 5](#), near the end of each DIMM socket. The DIMMs will partially eject from the socket.
 - b.) Remove the memory modules and place in an antistatic environment.

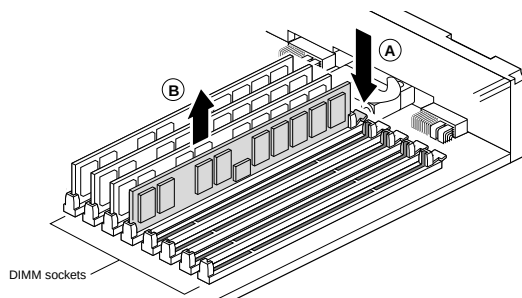


Figure 5 - DIMM Removal & Installation

2.2.4 ID Module

With patience, the ID module can be extracted from the frontplane without damage to either. If you find this process difficult, please see your system service manual on removing the ID module.

- 1.) Visually locate the ID module. It's on the frontplane near the top, as viewed looking into the slot vacated by the System Module. It is a small "silver" disk held in by a metal retaining clip.

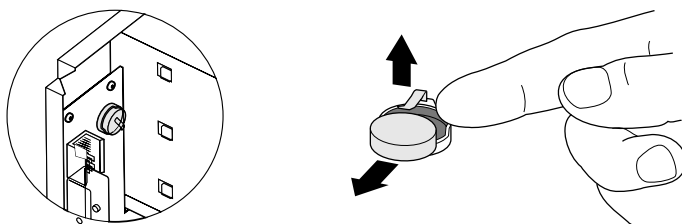


Figure 6 - System ID Module

- 2.) With one hand, reach inside the computer and use your fingers to extract the ID module.

2.3 Assemble Replacement Computer with Components



NOTICE
Potential for
equipment
damage

Do not use excessive force to install any component. Excessive force can result in component failure. Be patient and use only gentle but firm pressure.

2.3.1 ID Module

- 1.) Remove the system module from the replacement computer.
 - a.) Loosen the captive screw in each sliding handles.
 - b.) Pull both handles at the same time until they are fully extended.
 - c.) Pull the system module out.
 - d.) Place the system module on an antistatic surface component side up.
- 2.) Install ID Module into replacement computer
 - a.) With one hand, reach inside the replacement computer and use your fingers to insert the ID module.
 - b.) Visually inspect the ID module and make sure its securely in place.

2.3.2 Memory

- 1.) Install Memory Modules
 - a.) Insert the DIMMs onto the new system module and into their sockets. You'll hear a click when they are latched-in. Notice how the latch on the end of the socket moves up when they're seated properly. *DIMMs are notched on the bottom so that they cannot be inserted incorrectly.* See [Figure 5](#).

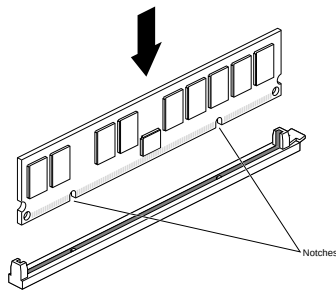


Figure 7 - Inserting Memory Modules

- b.) Verify that both sockets in a bank are populated. DIMMs must be installed in pairs. See [Figure 8](#).

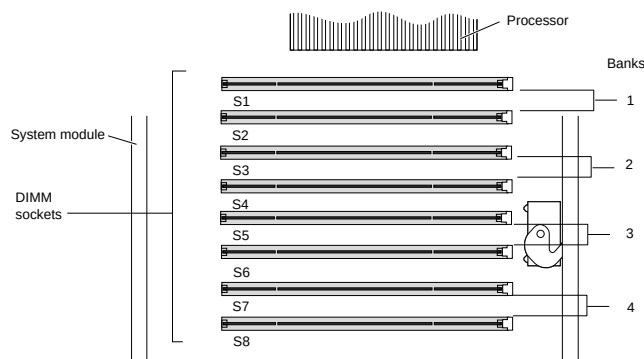


Figure 8 - DIMM Socket and Bank Identifications

- 2.) Install System Module back into replacement computer
 - a.) Grasp the module and push inwards, without allowing the delicate connectors on its back edge to touch anything.
 - b.) With the module inserted, push both handles in at the same time until they are fully seated.

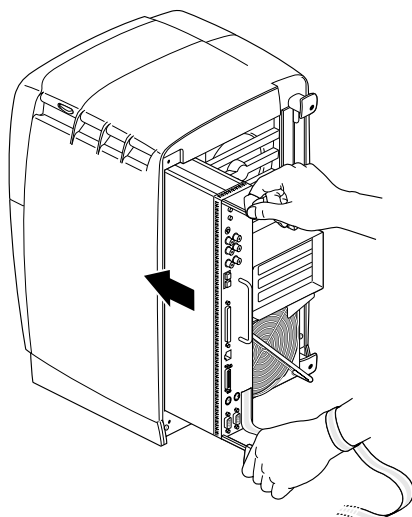


Figure 9 - System Module

- c.) Tighten the captive screw in each "sliding" handles.

2.3.3 PCI Module

- 1.) Slide the PCI module in the replacement computer. Do not to allow the compression connector to touch anything. ([Figure 3](#)).
- 2.) Push the release lever in to close (latch). Lever located along the bottom edge of the module ([Figure 10](#)).

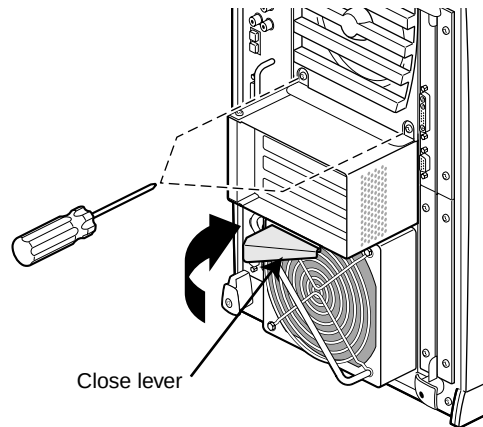


Figure 10 - PCI Module Installation

- 3.) Tighten the two captive screws (Figure 10).

2.3.4 Disk Drives

- 1.) Remove the replacement computer front bezel (cover). Pressing both bezel release buttons simultaneously to release the cover.
- 2.) Tilt the bezel toward you and lift up to remove.
- 3.) Grasp a drive handle and lift it to a horizontal position.
- 4.) Applying gentle and steady pressure, push the drives into place.
- 5.) Rotate the drive handle to the vertical position to lock the drive in place.
- 6.) Re-install the front bezel.

2.4 Install Replacement Computer in Console

2.4.1 Computer

- 1.) Place computer on computer tray.
- 2.) Reconnect cables.

2.4.2 Console Front Cover

- 1.) Install console's front cover.
 - a.) Lift the cover and engage the bottom edge of the cover with the bottom lip of the console.
 - b.) Slowly lean the cover inward until in place.
- 2.) Turn on system power
- 3.) Turn on console power switch to start-up computer

“END PROCEDURE”



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